

TIANAO LI

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📍 Department of Computer Science, McCormick School of Engineering, Northwestern University, Evanston, IL 60208

EDUCATION

Northwestern University

Ph.D. Candidate in Computer Science

GPA: 3.97/4.0

Research advisor: Emma Alexander

Evanston, IL

Sept. 2023 - Present

Northwestern University

M.S. in Computer Science

Evanston, IL

Sept. 2023 - Jun. 2026

Tsinghua University

B.Eng. in Electronic Engineering, Magna Cum Laude

Curricular Certificate in Astronomy

GPA: 3.85/4.0 (top 15%)

Research advisor: Cheng Ma, Lu Fang

Beijing, P.R. China

Aug. 2019 - Jun. 2023

HONORS & AWARDS

- Best Paper Honorable Mention, ICCV 2025 Workshop on Responsible Imaging *Oct. 2025*
- Best Poster Honorable Mention, Midwest Machine Learning Symposium (MMLS), 2025 *Jun. 2025*
- Outstanding Graduate, Tsinghua University (top 10%) *Jun. 2023*
- Scholarship of Comprehensive Excellence, Tsinghua University (top 10%) *Oct. 2022*
- Scholarship of Comprehensive Excellence, Tsinghua University (top 10%) *Oct. 2021*
- Scholarship of Social Work, Tsinghua University *Oct. 2020*

RESEARCH INTERESTS

My research interest is in the field of computational imaging and photography, which lies at the intersection of optics, signal processing, computer vision, and machine learning. Specifically, I study how generative priors, scene representations, and physics-based forward modeling can advance imaging — turning ill-posed measurements into faithful reconstructions of the world. My current applications span astronomical imaging, medical imaging, dynamic scene reconstruction, and monocular depth estimation.

PUBLICATIONS

* Equal contribution.

- [1] **Tianao Li**, Tjitske Starkenburg, Yu Sun, Emma Alexander. “**ShuffleFlow: Scalable Posterior Inference for Bayesian Inverse Imaging.**” *IEEE International Conference on Computational Photography (ICCP)*, 2026. **Best Poster Honorable mention**, *Midwest Machine Learning Symposium (MMLS)*, 2025.
- [2] Marcos A. Ferreira*, **Tianao Li***, John Mamish*, Josiah Hester, Yaman Sangar, Qi Guo, Emma Alexander. “**SpiderCam: Low-Power Snapshot Depth from Differential Defocus.**” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026 (**Highlight**).
- [3] **Tianao Li**, Manxiu Cui, Cheng Ma, Emma Alexander. “**Coordinate-based Speed of Sound Recovery for Aberration-Corrected Photoacoustic Computed Tomography.**” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. **Best Paper Honorable mention**, *ICCV 2025 Workshop on Responsible Imaging*.
- [4] **Tianao Li**, Emma Alexander. “**Galaxy Image Deconvolution for Weak Gravitational Lensing with Unrolled Plug-and-Play ADMM.**” *Monthly Notices of the Royal Astronomical Society: Letters*, 2023.

EXPERIENCE

Advanced Technology Group, Dolby Laboratories

PhD Research Intern

Jun. 2026 - Sept. 2026

Mentor: Jonathan Miller, Dongdong Fu

NSF-Simons AI Institute for the Sky (SkAI Institute)

Graduate Student Researcher

Jan. 2025 - Present

Advisor: Emma Alexander, Tjitske Starkenburg

Bio-Inspired Vision Lab, Northwestern University

Graduate Research Assistant

Sept. 2023 - Present

Advisor: Emma Alexander

Biophotonics Lab, Tsinghua University

Undergraduate Research Assistant

Oct. 2022 - Jun. 2023

Advisor: Cheng Ma

Bio-Inspired Vision Lab, Northwestern University

Research Intern (remote)

Apr. 2022 - Feb. 2023

Advisor: Emma Alexander

SIGMA Lab, Tsinghua University

Research Assistant

Sept. 2021 - Feb. 2022

Advisor: Lu Fang

INVITED TALKS

ShuffleFlow: Scalable Posterior Inference for Bayesian Inverse Imaging

IEEE International Conference on Computational Photography (ICCP), 2026

Jul. 2026

Coordinate-based Speed of Sound Recovery for Aberration-Corrected Photoacoustic Computed Tomography

ICCV 2025 Workshop on Responsible Imaging

Oct. 2025

Probabilistic Imaging of Galaxies for Weak Gravitational Lensing

Open SkAI 2025

Sep. 2025

Galaxy Image Deconvolution for Weak Gravitational Lensing with Unrolled Plug-and-Play ADMM

ECE Seminar, Department of Electrical & Computer Engineering, Boston University

Jul. 2024

Center for Interdisciplinary Exploration and Research in Astrophysics, Northwestern University

Jan. 2024

PKU Computational Scientific Imaging Group, Peking University

Dec. 2023

Astro Imaging Workshop 2023, Northwestern University

Jul. 2023

ACADEMIC SERVICE

- Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)
- Reviewer, British Machine Vision Conference (BMVC) 2026
- Reviewer, IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2026
- Reviewer, IEEE Open Journal of Signal Processing (OJSP)
- Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025